CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · DPP SHEET

Chapter 15: Similarity (With Applications to Maps and Models)

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. Two triangles have sides 4 cm, 6 cm, 8 cm and 6 cm, 9 cm, 12 cm. Are they similar? Name the condition.

2. $\triangle ABC \sim \triangle PQR$ with $AB = 5$ cm, $BC = 7$ cm, $PQ = 7.5$ cm. Find QR.

3. In $\triangle ABC$, $DE \parallel BC$ with $AD = 4$ cm, $DB = 6$ cm and $AE = 5$ cm. Find EC.

4. In $\triangle ABC$, $DE \parallel BC$ with $AD = 2$ cm, $AB = 6$ cm and $DE = 3$ cm. Find BC.

5. The corresponding sides of two similar triangles are in the ratio $3 : 5$. Find the ratio of their areas and of their perimeters.

6. The areas of two similar triangles are 49 cm² and 100 cm². If an altitude of the first is 3.5 cm, find the corresponding altitude of the second.

7. In $\triangle ABC$, D and E are the mid-points of AB and AC. Find the ratio of the area of $\triangle ADE$ to the area of $\triangle ABC$.

8. A model of a building is made to a scale of $1 : 50$. The model is 60 cm high. Find the height of the building.

9. For the model in the previous question, the floor area of a room in the model is 200 cm². Find the actual floor area in m².

10. A map has a scale of $1 : 50{,}000$. Two towns are 8 cm apart on the map. Find the actual distance in km, and the actual area in km² of a lake that covers 2 cm² on the map.

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Answer Key

1. Yes, SSS: each ratio is $\frac{2}{3}$
2. 10.5 cm
3. 7.5 cm
4. 9 cm
5. Areas $9:25$; perimeters $3:5$
6. 5 cm
7. $1:4$
8. 30 m
9. $200 \times 2500 = 500{,}000$ cm² $= 50$ m²
10. 4 km; $2 \times 0.25 = 0.5$ km²

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